

ANSWER KEY							AT# 10 (CAPACITOR)		
Q.	1	2	3	4	5	6	7	8	9
A.	2	1	2	3	1	3	2	3	1
Q.	10	11	12	13	14	15	16	17	18
A.	2	4	1	4	4	1	2	2	4
Q.	19	20	21	22	23	24	25	26	27
A.	2	3	4	4	4	1	1	4	4
Q.	28	29	30	31	32	33	34	35	36
A.	2	2	1	1	4	4	2	1	3
Q.	37	38	39	40	41	42	43	44	45
A.	4	2	1	4	3	4	3	3	2



Hints And Solution

1. ANS - 2

Energy of a charged capacitor, $E = \frac{1}{2} \frac{Q^2}{C}$

For a cylindrical capacitor, $C = \frac{2\pi\epsilon_0 L}{\log_e \left(\frac{b}{a} \right)}$

$$E = \frac{1}{2} \frac{Q^2}{2\pi\epsilon_0 L} \log_e \left(\frac{b}{a} \right)$$

Where L = length of the cylinder

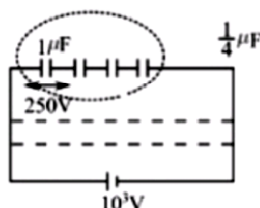
a and b = radii of two concentric cylinder

$$C' = \frac{2\pi\epsilon_0 (2L)}{\log_e \left(\frac{b}{a} \right)}$$

$$E' = \frac{1}{2} \frac{(2Q)^2}{C'} \frac{1}{2} \frac{(2Q)^2}{2\pi\epsilon_0 (2L)} \log_e \left(\frac{b}{a} \right)$$

$$E' = 2E$$

2. ANS - 1



Potential difference across each capacitor

$$V = \frac{1000}{n}$$

$$\frac{1000}{n} = 300$$

$$\therefore n = \frac{10}{3} \approx 4$$

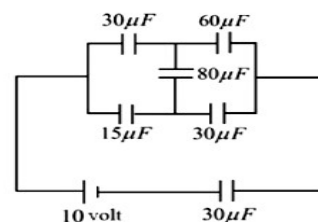
$$Eq = \frac{C}{n} m$$

$$\frac{1}{n} \times m = 2$$

$$m = n \times 2 = 4 \times 2 = 8$$

$$\therefore \text{minimum number of capacitors} = 8 \times 4 = 32$$

3. ANS - 2



The circuit shown in the figure is a balance Wheatstone bridge.

So the $80\mu\text{F}$ doesn't have any effect on it can be removed. After removing the $80\mu\text{F}$, we see that the $30\mu\text{F}$ and $60\mu\text{F}$ are in series.

Similarly, the $15\mu\text{F}$ and $30\mu\text{F}$ are also in series.

$$C_1 = \frac{30 \times 60}{30 + 60} = 20 \mu\text{F}, C_2 = \frac{15 \times 30}{15 + 30} = 10 \mu\text{F}$$

Hence, the equivalent capacitance of the entire circuit is

$$C_{eq} = C_1 + C_2 = 30$$

$$C_{eq} = 30/2 = 15\mu\text{F}$$

The charge that flows through the battery
 $= 10 \times 15 = 150 \mu\text{C}$

This charge will be divided in two branches in the ratio of capacitances.

On solving we get that charge on
 $60\mu\text{F} = 100 \mu\text{C}$

4. ANS - 3

As the charge remain constant, hence

$$Q = CV = C_1 V_1$$

$$Q = \frac{\epsilon_0 A}{d} V = \frac{\epsilon_0 A}{d - t + \frac{1}{k}} V_1$$

$$\frac{\epsilon_0 A}{0.005} (25) = \frac{\epsilon_0 A}{0.002 + \frac{0.003}{k}} (V)$$

$$V = 11.5 \text{ V}$$

5. ANS - 1

During charging of capacitor, charge at any instant is

$$q = q_0 (1 - e^{-\frac{t}{RC}})$$

$$0.99q_0 = q_0 (1 - e^{-\frac{t}{RC}})$$

$$0.99 = (1 - e^{-\frac{t}{RC}})$$

$$0.01 = e^{-\frac{t}{RC}}$$

$$e^{\frac{t}{RC}} = 100$$

$$t = RC \ln 100 = 2 \times 100 \times 10^{-6} \times 2 \times 2.303 = 0.92 \text{ ms}$$

6. ANS - 3



Capacitance does not depend on charge on it,

$$C = \frac{\epsilon_0 A}{d}$$

In final charge distribution, charge on outer plates $q_1 = q_4 = q$ and on inner plates

$$q_2 = q_3 = 0$$

As charge on inner plates are zero so there is no field between the plates and no potential difference and no energy is stored.

7. ANS - 2

Resistance in uppermost branch

$$R = \frac{2 \times 3}{2 + 3} = \frac{6}{5} = 1.2 \Omega$$

After a long time, branch with the capacitor can be considered as open circuited

Hence

$$R_{eq} = 2.8 + 1.2 = 4$$

$$I = \frac{V}{R} = \frac{6}{4} = \frac{3}{2} \text{ A}$$

$$I_{2\Omega} = \frac{IR_2}{R_1 + R_2} = \frac{\frac{3}{2} \times 3}{5} \text{ A} = 0.9 \text{ A}$$

8. ANS - 3

We know for a parallel plate capacitor,

$$F = \frac{Q^2}{2A\epsilon_0}$$

Since the capacitor is isolated charge will not change, hence force will remain unchanged

9. ANS - 1

$$\begin{aligned}
 C_{\text{parallel}} &= \frac{25}{6} C_{\text{series}} \\
 \Rightarrow C_1 + C_2 &= \frac{25}{6} \frac{C_1 C_2}{C_1 + C_2} \\
 \Rightarrow 6(C_1 + C_2)^2 &= 25 C_1 C_2 \\
 \Rightarrow 6C_1^2 + 6C_2^2 + 12C_1 C_2 &= 25 C_1 C_2 \\
 \Rightarrow 6 \frac{C_1^2}{C_2^2} + 6 + 12 \frac{C_1}{C_2} &= 25 \frac{C_1}{C_2} \\
 \Rightarrow 6 \frac{C_1^2}{C_2^2} - 13 \frac{C_1}{C_2} + 6 &= 0
 \end{aligned}$$

This is quadratic equation, By solving we get

$$\frac{C_2}{C_1} = \frac{3}{2}$$

10. ANS - 2

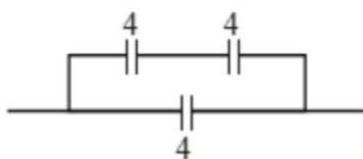
During discharging of a capacitor, the current is given by,

$$\begin{aligned}
 I &= \frac{E}{R} e^{-\frac{t}{RC}} \\
 \log I &= \log \frac{E}{R} e^{-\frac{t}{RC}} \\
 \log I &= -\frac{t}{RC} + \log \frac{E}{R}
 \end{aligned}$$

Intercept is constant = E & R constant

Magnitude of slope decreases = C is increased

11. ANS - 4



For top branch,

$$\begin{aligned}
 \frac{1}{C_{\text{eq}}} &= \frac{1}{4} + \frac{1}{4} \\
 \Rightarrow C_{\text{eq}} &= 2\Omega
 \end{aligned}$$

For parallel combination,

$$C_{\text{net}} = 2\Omega + 4\Omega = 6\Omega$$

12. ANS - 1

Initially

$$C_1 = 2C \quad \text{and} \quad C_2 = C$$

$$q_1 = 2CV$$

$$q_2 = CV$$

Now condenser of capacity C is filled with dielectric K, therefore

$$C'_2 = KC$$

As charge is conserved

$$q_1 + q_2 = (C'_2 + 2C)V'$$

$$V' = \frac{3V}{K + 2}$$

13. ANS - 4

Charge on 6 μF capacitor = charge on 12 μF capacitor.

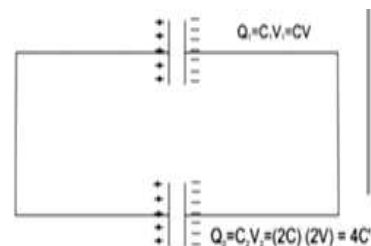
$$6 \times 10^{-6} \times 2 \text{ V} = 12 \times 10^{-6} \text{ C}$$

So, Potential difference across 12 μF capacitor is

$$V = \frac{12 \times 10^{-12}}{12 \times 10^{-1}} = 1 \text{ V}$$

$$\text{Battery voltage} = 2 \text{ V} + 1 \text{ V} = 3 \text{ V}$$

14. ANS - 4



Diagrammatic representation of given problem is shown in figure. The net charge shared between the two capacitors is, The two capacitors will have the same potential Say V' . The net capacitance of The parallel combination of the two capacitor will be

$$C' = C_1 + C_2 = C + 2C = 3C$$

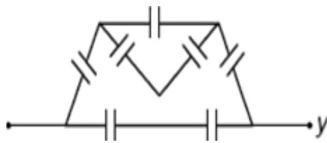
The potential difference across the capacitors will be

$$V' = \frac{Q'}{C'} = \frac{3C}{CV} = V$$

The electrostatic energy of the capacitors will be

$$U' = \frac{1}{2} C' V'^2 = \frac{1}{2} (3C) V^2 = \frac{3}{2} C V^2$$

15.ANS - 1



As the circuit has left-right symmetry, we can disconnect as shown in the figure.

16.ANS - 2

In a capacitor, energy is stored in the electric field between the plates.

$$\begin{aligned} \Delta U &= U_f - U_i \\ &= \frac{1}{2} C V_f^2 - \frac{1}{2} C V_i^2 = \frac{1}{2} C (V_f^2 - V_i^2) \end{aligned}$$

$$\begin{aligned} \text{Given } C &= 6 \mu F = 6 \times 10^{-6} F \\ V_i &= 10V \\ V_f &= 20V \end{aligned}$$

$$\begin{aligned} \therefore \Delta U &= \frac{1}{2} \times 6 \times 10^{-6} ((20)^2 - (10)^2) \\ &= 3 \times 10^{-6} \times 300 \\ &= 9 \times 10^{-4} J \end{aligned}$$

17.ANS - 2

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

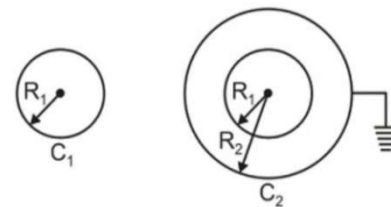
18.ANS - 4

$$\text{The energy stored in the capacitor} = \frac{1}{2} C V^2 =$$

$$\frac{1}{2} \times 4 \times 10^{-6} \times 400 \times 400 = 0.32 J$$

This energy will be converted into heat in the resistor.

19.ANS - 2



$$C_1 = 4\pi\epsilon_0 R_1$$

$$\text{And } C_2 = 4\pi\epsilon_0 \left(\frac{R_1 R_2}{R_2 - R_1} \right)$$

$$\text{Given that } C_2 = n C_1$$

$$\frac{R_1 R_2}{R_2 - R_1} = n R_1$$

$$\frac{R_2 / R_1}{R_2 - R_1} = n$$

$$\frac{R_2}{R_1} = \frac{n}{n - 1}$$

20.ANS - 3

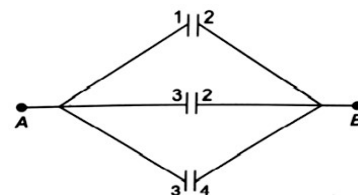


Plate 1 is connected to plate 3 and plate 2 is connected to plate 4.

Thus, there are three capacitors in parallel, each of capacitance

$$C = \frac{\epsilon_0 A}{d}$$

The equivalent capacitance, as shown in the figure, is

$$C' = 3C = \frac{3\epsilon_0 A}{d}$$

21. ANS - 4

Here distance between parallel plates $d = 4\text{mm} = 0.004\text{ m}$ and $k = 3$, thickness $t = 3\text{mm} = 0.003\text{ m}$

$$C = \frac{\epsilon_0 A}{d} \text{ and } C_1 = \frac{\epsilon_0 A}{d_1 - t(1 - \frac{1}{k})}$$

$$\text{Since } C_1 = \frac{2}{3}C$$

$$\frac{\epsilon_0 A}{d_1 - t(1 - \frac{1}{k})} = \frac{2}{3} \frac{\epsilon_0 A}{d}$$

$$\frac{1}{d_1 - 0.003(1 - \frac{1}{3})} = \frac{2}{3 \times 0.004}$$

$$\frac{1}{d_1 - 0.003 \times \frac{2}{3}} = 1/0.006$$

$$\frac{1}{d_1 - 0.002} = \frac{1}{0.006}$$

$$d_1 - 0.002 = 0.006$$

$$d_1 = 0.002 + 0.006 = 0.008\text{ m} = 8\text{mm}$$

22. ANS - 4

$$E(\text{initial}) = \frac{Q_0^2}{2C} = \frac{1}{2} CV_0^2 = V_0^2$$

$$\text{Final energy } E(f) = \frac{Q_0^2}{2C_{\text{eff}}}$$

$$E(f) = \frac{1}{2} \times \frac{2v_0}{10} \times 2v_0 = \frac{v_0^2}{5}$$

$$\Delta E(\text{loss}) = \frac{4v_0^2}{5}$$

$$\% \text{loss} = \frac{4v_0^2}{5v_0^2} \times 100$$

$$\Delta E(\text{loss})\% = 80\%$$

23. ANS - 4

Decrease in energy of the system after connection is given by

$$\Delta U = \frac{1}{2} \left(\frac{C_1 C_2}{C_1 + C_2} \right) (V_1 - V_2)^2$$

$$\Delta U = 0, \text{ if } V_1 = V_2$$

$$\text{or } \frac{kq_1}{R_1} = \frac{kq_2}{R_2}$$

$$\text{or } q_1 R_2 = q_2 R_1$$

24. ANS - 1

$$C_1 = 10\text{ }\mu\text{F} ; C_2 = ?$$

$$V_1 = 100\text{V} ; V_2 = 0$$

$$V_{\text{common}} = 40\text{V}$$

$$40 = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$40 = \frac{10 \times 100 + 0}{10 + C_2}$$

$$40C_2 = 1000 - 400$$

$$C_2 = 15\text{ }\mu\text{F}$$

25. ANS - 1

$$\text{New capacitance } C = KC_0$$

$$\text{New electric field, } E = \frac{E_0}{K}$$

$$\text{New potential difference } \Delta V = \frac{Q_0}{C} = \frac{Q_0}{KC_0} = \frac{\Delta V_0}{K}$$

26. ANS – 4

The capacitance of a parallel plate condenser is,

$$C = \frac{\epsilon_0 k A}{d}$$

Therefore capacitance does not depend on the metal of the plates.

27. ANS – 4

For charge Q_1 , the electric field is

$$E_1 = \frac{Q_1}{2\epsilon_0 A}$$

where A is the area, For charge Q_2 , the electric field is

$$E_2 = \frac{Q_2}{2\epsilon_0 A}$$

Resultant electric field $E = E_1 - E_2 = \frac{Q_1 - Q_2}{2\epsilon_0 A}$

The potential difference between plates when they are brought close together to form a parallel plate capacitor is

$$V = Ed = \frac{Q_1 - Q_2}{2\epsilon_0 A} d$$

Since $C = \frac{\epsilon_0 A}{d}$ for parallel plate capacitor

$$\therefore V = \frac{Q_1 - Q_2}{2C}$$

28. ANS – 2

Dielectric strength means the maximum electric field which a medium can bear. Here, if field becomes more than this, then the charge will start leaking from the metal ball.

$$\begin{aligned}
 \text{So, } E &= \frac{kQ}{R^2} \Rightarrow Q = \frac{ER^2}{k} \\
 &= \frac{3 \times 10^6 \times (0.001)^2}{9 \times 10^9} = \frac{1}{3} \text{ nC}
 \end{aligned}$$

29. ANS – 2

$$U_i = \frac{1}{2} C \left(\frac{V}{3} \right)^2 + \frac{1}{2} C \left(\frac{V}{3} \right)^2 + \frac{1}{2} C \left(\frac{2V}{3} \right)^2$$

$$U_f = \frac{1}{2} CV^2$$

$$W_b = \frac{CV^2}{3}$$

$$\Delta H = [W_b - (U_f - U_i)] = \frac{CV^2}{6} = 0.30 \text{ mJ}$$

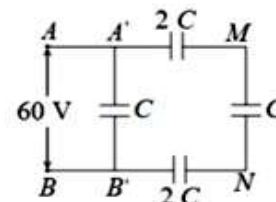
30. ANS – 1

$$C_{eq} = \frac{2C_0}{3} + \frac{4C_0}{3} = 2C_0$$

31. ANS – 1

Since the plates are being moved very slowly there is no heat loss. So the total work done by external agent and battery will be equal to the change in potential energy of the system.

32. ANS – 4



Equivalent capacitor in circuit A'MNB' is $\frac{C}{2}$.

Charge on capacitor C between MN is $60 \times \frac{C}{2} = 30C$

$$\therefore V_{MN} = \frac{30C}{C} = 30V$$

33. ANS – 4

ΔU = decrease in potential energy
 $= U_i - U_f$

$$= \frac{1}{2} C(V_1^2 + V_2^2) - \frac{1}{2} (2C) \left(\frac{V_1 + V_2}{2} \right)^2$$

$$= \frac{1}{4} C(V_1 - V_2)^2$$

34. ANS - 2

$$\text{Power, } p = \frac{\text{energy}}{\text{time}} = \frac{\frac{1}{2}cv^2}{t}$$

$$P = \frac{CV^2}{2t}$$

$$P = \frac{40 \times 16^{-6} \times 9 \times 10^6}{2 \times 2 \times 10^{-3}} \text{ W}$$

$$P = 90 \times 1000$$

$$P = 90 \text{ kW}$$

35. ANS - 1

We know that spring force $F=kx$ will balance electrostatic force of attraction. Therefore,

$$\frac{q^2}{2A\epsilon_0} = kx$$

$$x = \frac{q^2}{2A\epsilon_0 K}$$

36. ANS - 3

The potential difference E_0 is given to both of the capacitor. The two capacitors are then separately allowed to discharge through same resistors.

Since both capacitors are charged to same potential difference hence current at $t = 0$, $I_0 = \frac{E_0}{R}$ will be same in both circuits.

Time taken in 50% discharging = $RC \ln(2)$

As 1st capacitance is 50% less than that of 2nd capacitance, Hence 1 μF will take less time compared to 2 μF .

37. ANS - 4

$$E = \left(\frac{1}{2}\right) CV^2 \dots (i)$$

The energy stored in capacitor is lost in the form of heat energy

$$H = ms\Delta T \dots (ii)$$

From eq (i) and (ii), we have

$$ms\Delta T = \left(\frac{1}{2}\right) CV^2$$

$$V = \sqrt{\frac{2ms\Delta T}{C}}$$

38. ANS - 2

In the direction parallel to plate

$$\Rightarrow v_1 \cos \alpha = v_2 \cos \beta \Rightarrow \frac{v_1}{v_2} = \frac{\cos \beta}{\cos \alpha}$$

ratio of kinetic energy

$$\Rightarrow \frac{KE_1}{KE_2} = \left(\frac{v_1}{v_2}\right)^2 \Rightarrow \frac{KE_1}{KE_2} = \left(\frac{\cos \beta}{\cos \alpha}\right)^2$$

39. ANS - 1

The two condensers with K and with air are in parallel. With air,

$$C_1 = \frac{\epsilon_0}{d} \left(\frac{3A}{4}\right) = \frac{3\epsilon_0 A}{4d}$$

With medium,

$$C_2 = \frac{\epsilon_0 K}{d} \left(\frac{A}{4}\right) = \frac{\epsilon_0 AK}{4d}$$

So,

$$C' = C_1 + C_2$$

$$C' = \frac{3\epsilon_0 A}{4d} + \frac{\epsilon_0 AK}{4d} = \frac{\epsilon_0 A}{d} \left[\frac{3}{4} + \frac{K}{4}\right]$$

$$C' = \frac{C}{4} (K + 3) \left[\because C = \frac{A\epsilon_0}{d}\right]$$

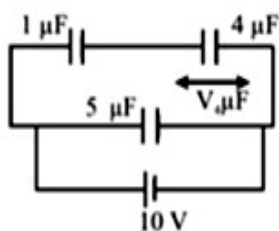
40. ANS - 4

$$\text{Loss of energy} = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \frac{5 \times 10^{-6} \times 5 \times 10^{-6} (2000 - 1000)^2}{(5 \times 10^{-6}) + (5 \times 10^{-6})}$$

$$= \frac{5}{4} = 1.25 \text{ J}$$

41. ANS - 3



$3\mu\text{F}$ and $1\mu\text{F}$ are in parallel hence

$C_{\text{net}} = C_1 + C_2$, Hence it is equal to $4\mu\text{F}$

Now $2\mu\text{F}$ & $2\mu\text{F}$ are in series

With this equivalent capacitor ($4\mu\text{F}$).

Solving $2\mu\text{F}$ and $2\mu\text{F}$ are in series, we get $1\mu\text{F}$.

$$\therefore V_{4\mu\text{F}} = \frac{V \times C_{1\mu\text{F}}}{C_{1\mu\text{F}} + C_{4\mu\text{F}}}$$

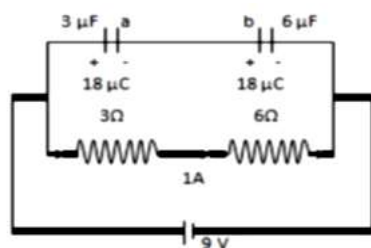
$$\Rightarrow V_{4\mu\text{F}} = \frac{10 \times 1}{(1 + 4)} \Rightarrow V_{4\mu\text{F}} = 2V$$

(Potential difference across $4\mu\text{F}$ capacitor is calculated by voltage division rule)

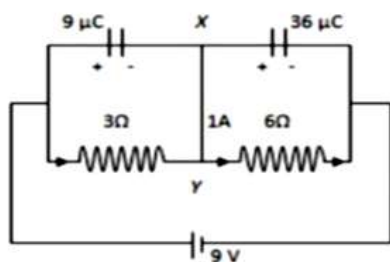
$$\therefore V_{5\mu\text{F}} = 10V \quad (\text{Potential difference across } 5\mu\text{F} \text{ is same as battery})$$

$$\therefore \frac{V_{1\mu\text{F}}}{V_{5\mu\text{F}}} = \frac{2}{10} = \frac{1}{5}$$

42. ANS - 4



Initial Figure
(when switch was open)

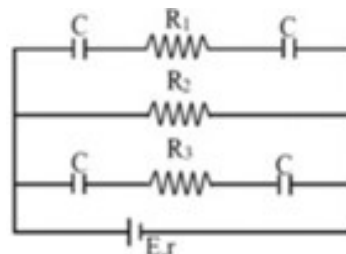


Final Figure
(when switch is closed)

From Y to X charge flows to plates a and b.

$$(q_A + q_B)_i = 0, (q_A + q_B)_f = 27\mu\text{C}$$

43. ANS - 3



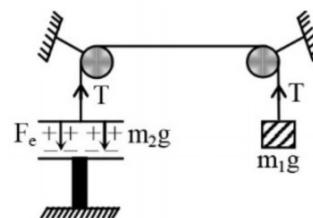
Current through resistors R_1 and R_3 in steady state is zero. Therefore current through R_2 is

$$i_2 = \frac{E}{R_2 + r} = \frac{5}{4 + 1} = 1A$$

Therefore charge on capacitor is

$$q = C \left(\frac{i_2 R_2}{2} \right) = 6\mu\text{C}$$

44. ANS - 3



$$T = Fe + m_2g \dots \dots (1) (\text{for upper plate})$$

$$T = m_1g \dots \dots (2) (\text{for } m_1)$$

$$\therefore m_1g = \frac{q^2}{2A\epsilon_0} + m_2g$$

$$\Rightarrow m_1 = \frac{q^2}{2A\epsilon_0g} + m_2$$

45. ANS - 2

As battery is disconnected, so charge will remain the same. It is given that final potential is the same. So final capacitance should be $C_1 = C_2$

$$\frac{A\epsilon_0}{d} = \frac{A\epsilon_0}{(1.6 + d) - t \left(1 - \frac{1}{K} \right)} \Rightarrow K = 5$$